Ryan Maxin

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EDUCATION

University of Waterloo **Sep 2021 – Aug 2026 (expected)**
Candidate for Bachelor of Computer Science, Co-op ***(GPA: 3.9)*** *Waterloo, Ontario*

EXPERIENCE

NVIDIA **Sept 2025 – Present**
GPU Compiler Engineer Intern *Austin, Texas*

- Contributing to GPU back-end compiler **optimizations** for NVIDIA's **SIMT** architecture, focusing on code generation, performance tuning, and parallel execution analysis in **C++**.

Qualcomm **May 2025 – Aug 2025**
ML Compiler Engineer Intern *Austin, Texas*

- Architected a sharding algorithm for transformer-based **LLMs** in **PyTorch**, partitioning **MLP** and **attention** blocks across a fixed number of **NSP cores** to enhance **parallel execution** and **enable scalable on-device inference**.
- Lowered the sharded graph into the NSP compiler, modifying its core-mapping logic in **C++** to apply mappings produced by the partitioning algorithm, achieving up to a **50% reduction** in inter-core communication overhead.
- Engineered custom **all-reduce/all-gather** primitives at the **PyTorch** level to balance synchronization across compute units, **maximizing core utilization**.

AMD **Jan 2025 – Apr 2025**
Software Engineer Intern *Toronto, Ontario*

- Implemented **Vulkan**, **DirectX**, **OpenGL**, and **OpenCL compute kernels** to enable cross-platform color-space and scaling conversions for **YUV 4:4:4** and **4:2:2** formats within AMD's Advanced Media Framework.
- Unified and optimized host-side conversion paths in **C++**, eliminating **~1,500** lines of redundant code and precomputing shader parameters to achieve a **~6%** runtime performance improvement.
- Built an automated testing pipeline validating **20+** format conversions across **1000s** of configuration combinations, performing statistical **SSIM/PSNR** analysis to ensure high-fidelity and consistent results.

Huawei **May 2024 – Aug 2024**
Compiler Engineer Intern *Toronto, Ontario*

- Enhanced loop tiling **compiler optimization** in **LLVM** using **C++**, successfully incorporating it as a default pass and boosting runtime performance of **high-performance computing** workloads by up to **7%**.
- Implemented an **LLVM** compiler pass in **C++** to optimize high-bandwidth memory allocation in **multithreaded** programs by leveraging **IR** pointer analysis, reducing memory contention by as much as **60%**.
- Integrated **profile-guided optimization** into the pass using cache miss data from the **perf** tool to further improve program throughput.

WSIB Innovation Lab **Sep 2023 – Dec 2023**
Compiler Engineer Intern *Waterloo, Ontario*

- Architected a **generative AI** tool to enable conversational interaction with **SQL** databases.
- Leveraged the unique strengths of different quantized and finetuned **LLMs** using **Python** and **LangChain** to achieve **parity with GPT-4** in database QA, effectively addressing OpenAI data privacy concerns.
- Deployed the AI to **Azure VMs** and presented the success to a **150-person audience** in the CBIA branch.

Infinera **Sep 2022 – Dec 2022**
Firmware Engineer Intern *Ottawa, Ontario*

- Created a CLI API in **C++** using **eRPC** to enable low-level runtime reconfiguration of digital subcarriers, **eliminating a 10-minute recompile time** for developers.

- Utilized the Atlassian API, **Pandas** and **Matplotlib** in **Python** to display real-time graphical data of production build data from **Jenkins** CI on a confluence page, allowing the team to identify and fix resource inefficiencies on hardware limited to 32MB of memory.

University of Waterloo **Jan 2022 – Apr 2022**
Course & Technical Support Assistant *Waterloo, Ontario*

- Engineered a **JavaScript** algorithm to process **1000s** of raw Sentinel-1 arctic satellite RADAR images in Google Earth Engine to determine melt/freeze onset dates for lakes and rivers in Canada.

PROJECTS

Sorting Algorithm Visualizer

- Developed an interactive visualizer of **six** popular sorting algorithms using **React** and **Material UI**.

Realm Tunes

- Created a feature-rich Discord music bot using **Python** and the Discord.py API.
- Deployed Realm Tunes to an **Ubuntu** server, providing on-demand music to **350 users** across **7 servers**.

SKILLS

Languages: C/C++, Python, JavaScript, TypeScript, Java, Kotlin, HTML/CSS, SQL, Assembly, VHDL
Technologies: Git, LLVM, Docker, Perf, PostgreSQL, SQLite, Firebase, Jenkins, Azure, Jira